

Cryptography

Unit- 2.3

Integer Arithmetic

- In integer arithmetic, we use set and few operations.

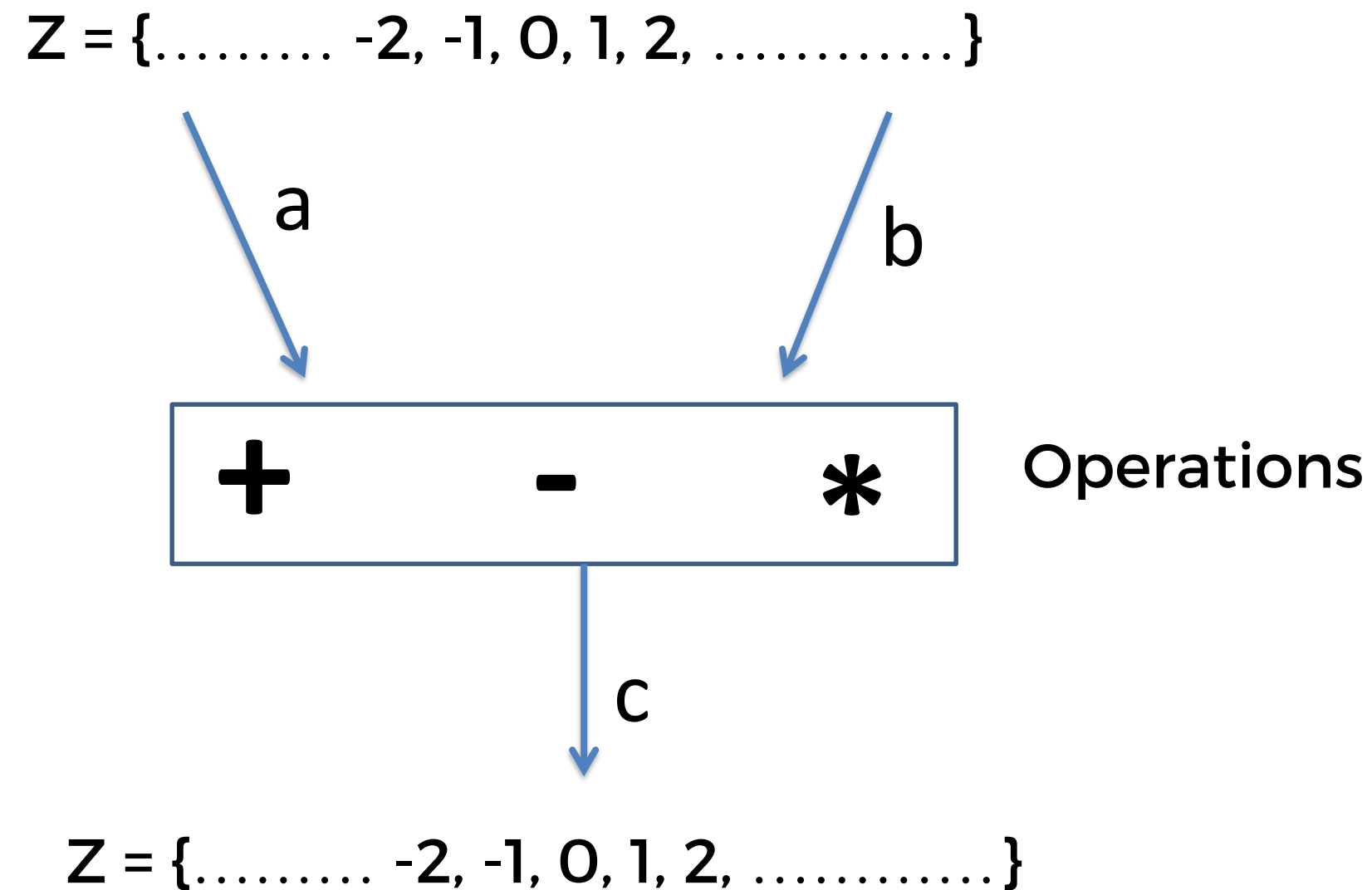
Set of Integers :

- The set of integers, denoted by Z , contains all integral number (with no fraction) from negative infinity to positive infinity.

$$Z = \{..... -2, -1, 0, 1, 2,\}$$

Binary Operation :

- In cryptography, we are interested in three binary operations applied to the set of integers. A binary operation takes two inputs and creates one output.



Integer Division

- In integer arithmetic, if we divide 'a' by 'n', we get 'q' and 'r'. The relationship between these four integers can be shown as-

$$a = q * n + r$$

Where, a → Dividend

q → Quotient

n → Divisor

r → Remainder

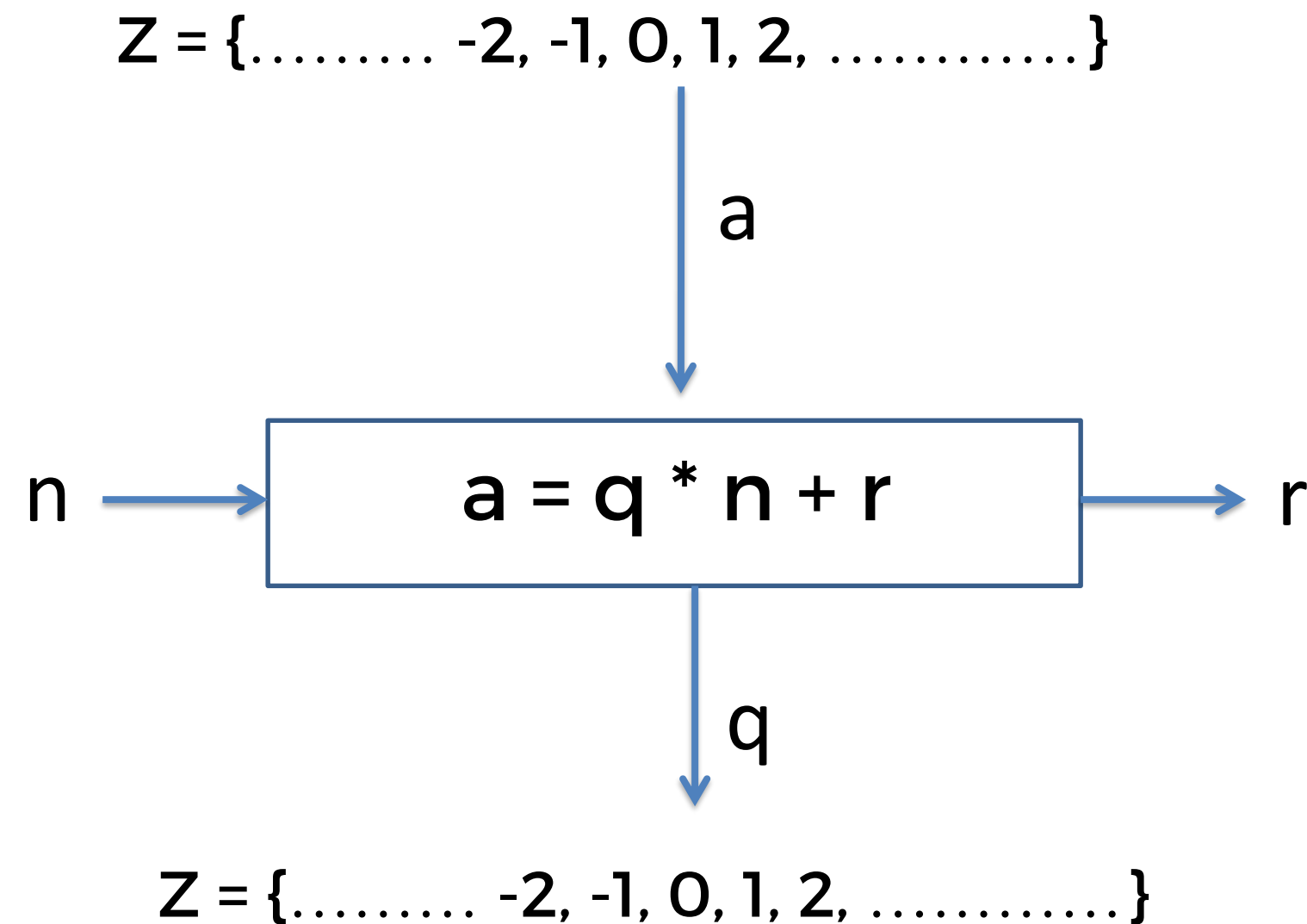
Diagram illustrating integer division:

$$\begin{array}{r} 23 \\ \hline 11 \overline{) 255} \\ \underline{-22} \\ 35 \\ \underline{-33} \\ 2 \end{array}$$

Labels: n (Divisor), q (Quotient), r (Remainder)

Two Restrictions on Division

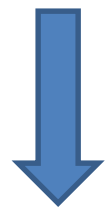
- For our purpose, we impose two restrictions.
- First, we require that the divisor be a positive integer ($n > 0$).
- Second, we require that the remainder be a non-negative integer ($r \geq 0$)



Two Restrictions on Division

- When we use calculator, r and q are negative when a is negative. Now **how we can apply the restriction that r needs to be positive?**
- Here the solution is simple, we decrement the value of q by 1 and we add the value of n to r to make it positive.

$$-255 = (-23) * 11 + (-2)$$



$$-255 = (-24) * 11 + 9$$

GCD (Greatest Common Divisor)

FACT-1 : The integer 1 has only one divisor, itself.

FACT-2 : Any +ve integer has at least two divisors, 1 and itself (but it can have more).

- The Greatest Common Divisor of two positive integers is the largest integer that can divide two integers.
- For example: $\text{GCD} (20, 40) = 20$

GCD (Greatest Common Divisor)

Euclidean Algorithm:

1. $\text{GCD}(a, 0) = a$

2. $\text{GCD}(a, b) = \text{GCD}(b, r)$

Where r is a remainder of dividing a by b .

Q) Find $\text{gcd}(36, 10)$ using Euclidean Algorithm.

Ans:

$$\begin{aligned}\text{Here, } \text{gcd}(36, 10) &= \text{gcd}(10, 6) \\ &= \text{gcd}(6, 4) \\ &= \text{gcd}(4, 2) \\ &= \text{gcd}(2, 0) \\ &= 2\end{aligned}$$

Note: When $\text{GCD}(a, b) = 1$, we say that a and b are relatively prime.

GCD (Greatest Common Divisor)

Euclidean Algorithm:

Algorithm:

$r_1 = a;$

$r_2 = b;$

While($r_2 > 0$)

{

$q = r_1 / r_2 ;$

$r = r_1 - q * r_2 ;$

$r_1 = r_2 ;$

$r_2 = r;$

}

$\text{gcd} (a, b) = r_1 ;$

r_1	r_2	q	r
36	10	3	6
10	6	1	4
6	4	1	2
4	2	2	0
2	0	-	-

$\text{gcd} (36, 10) = 2$

GCD (Greatest Common Divisor)

Q) Find gcd of 161 and 28.

Ans :

Here, $a = 161$

and $b = 28$

Now,

r_1	r_2	q	r
168	28	5	21
28	21	1	7
21	7	3	0
7	0	-	-

$\therefore \text{gcd}(161, 28) = 7$

Algebraic Structure

- The combination of the set and the operations that are applied to the element of the set is called an algebraic structure.
- The three common algebraic structures are –
 1. Groups
 2. Rings
 3. Fields

1. Groups

- Denoted as $\{G, .\}$ or $\langle G, . \rangle$
- A group 'G' is a set of element with a binary operation $(.)$ that satisfies the following 4 - properties.

A1] Closure Property:

If 'a' & 'b' are the element of 'G', then $c = a . b$ is also an element of G.
i.e. For all $a, b \in G$, then $a . b \in G$.

A2] Associativity Property:

If 'a', 'b' & 'c' are the element of 'G', then
 $(a . b) . c = a . (b . c)$
i.e. For all $a, b, c \in G$, then $(a . b) . c = a . (b . c)$

A3] Existence of Identity Property:

For all element 'a' in 'G', there exists an element 'e', called the identity element such that-
 $e . a = a . e = a$

1. Groups

A4] Existence of Inverse Property:

For each element 'a' in 'G', there exists an element 'a⁻¹', called the inverse of 'a' such that

$$a \cdot a^{-1} = a^{-1} \cdot a = e$$

Q] Show that $\langle \mathbb{Z}, + \rangle$ is a group .

Ans:

To show that $\langle \mathbb{Z}, + \rangle$ is a group, we need to verify that the set of integers $\mathbb{Z}$ with the operation of addition (+) satisfies the four group properties: closure, associativity, identity, and invertibility.

Here, $\mathbb{Z} = \{ \dots -2, -1, 0, 1, 2, \dots \}$

1. Closure Property :
2. Associativity Property :
3. Existence of Identity :
4. Existence of Inverse :

1. Groups

Q] Is $\langle \mathbb{Z}, * \rangle$ is a group ?

Ans:

Abelian Group

A group is called abelian group if it satisfies one extra property, commutativity.

A1] Closure

A2] Associativity

A3] Existence of identity

A4] Existence of Inverse

A5] Commutativity :

For all a, b in G , then

$$a \cdot b = b \cdot a$$

2. Rings

- Denoted as $\{R, +, *\}$ or $\langle R, +, * \rangle$
- Is an algebraic structure with two binary operation $+$, $*$.
- Here, first operation must satisfy all 5 properties of the Abelian group.

A1] Closure Property

A2] Associativity Property

A3] Existence of Identity Property

A4] Existence of Inverse Property

A5] Commutative Property

- The second operation must satisfy only these two properties-

M1] Closure Property

M2] Associativity Property

2. Rings

- Plus, the second operator should also be -

M3] Distributive :

- For all $a, b, c \in R$
 $a * (b + c) = (a * b) + (a * c)$
 $(a + b) \times c = (a \times c) + (b \times c)$

M4] Commutative:

- For all $a, b \in R$
 $a * b = b * a$

M5] Multiplicative Identity :

- $a * e = e * a = a$ for all $a \in R$

M6] No zero divisor :

- If $a * b = 0$ then either $a = 0$ or $b = 0$ for all $a, b \in R$

- If rule [A1 – M4] = Commutative Ring

- If rule [A1 – M6] = Integral domain

2. Rings

Q] Show that the set $\mathbb{Z}$ with two operations , addition and multiplication, is a commutative ring. i.e $\langle \mathbb{Z}, +, * \rangle$

3. Fields

- Denoted as $\{F, +, *\}$ or $\langle F, +, * \rangle$
- Is a set of elements with two binary operations addition and multiplication, such that following axioms are obeyed.
- [A1 – M6]
- **M7] Multiplicative Inverse :**
 $a * a^{-1} = a^{-1} * a = 1$

Finite field:

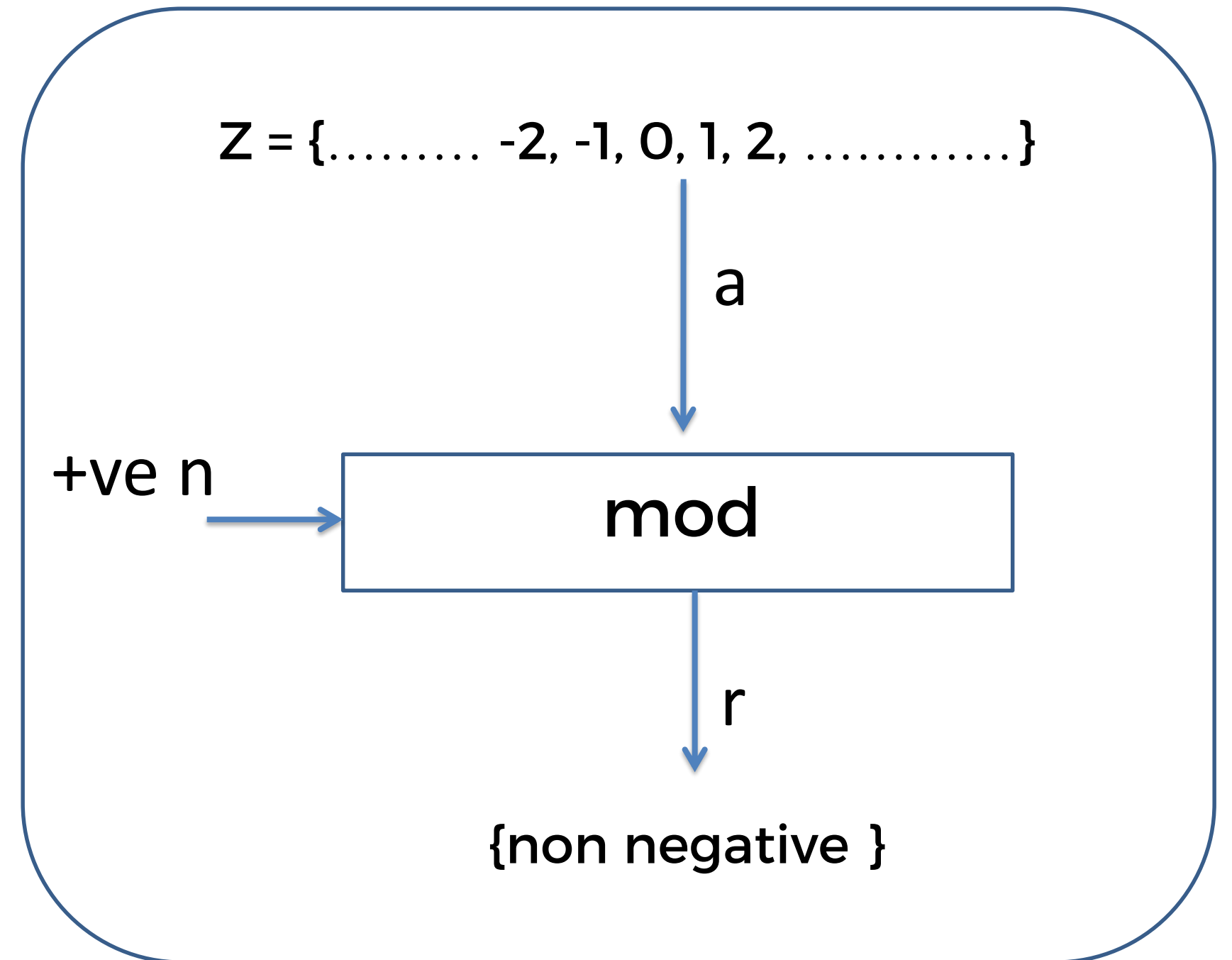
- Finite field is a field with a finite number of elements.
- Galois showed that for a field to be finite, the number of elements should be P^n , where P is a prime and n is any positive integer.
- Finite fields are usually called Galois Field and denoted as $Gf(P^n)$.
- Galois' Theorem:
- For a field to be finite, the number of elements must be P^n
 - where: P = a prime number (2, 3, 5, 7, 11...)
 - n = any positive integer (1, 2, 3...)

Modular Arithmetic

- The division relationship $(a = q * n + r)$ has two inputs $(a \text{ \& } n)$ and two outputs $(q \text{ \& } r)$.
- But, in modular arithmetic, we are interested in only one of the output, the remainder r ; we do not care about quotient q .

Modulo Operator :

- By using modulo operator, we want to know what is the value of r when we divide a by n .
- The modulo operator is shown as `mod`. Mod is also a binary operator with two input a & n and one output r .
- Here the input n is called the modulus and the output r is called the residue.
- Syntax: $a \bmod n = r$



Q] Solve:

1] $27 \bmod 5$

2] $36 \bmod 12$

3] $-18 \bmod 14$

4] $-7 \bmod 10$

Set of Residues

- The result of the modulus operation with modulus 'n' is always an integer between 0 and n-1.
- We can say that the modulo operation creates a set, which in modular arithmetic is referred to as +ve set of least residues modulo 'n' or Z_n .

$$Z_n = \{ 0, 1, 2, 3, \dots, (n-1) \}$$

- $Z_2 = \{ 0, 1 \}$
- $Z_6 = \{ 0, 1, 2, 3, 4, 5 \}$
- $Z_{11} = \{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 \}$

Congruence:

- In cryptography, we often use the concept of congruence instead of equality.
- To show that two integers congruent, we use the congruence operator ($\equiv$).

Example, 1) $2 \equiv 12 \pmod{10} \Rightarrow 2 \bmod 10 = 12 \bmod 10$

2) $3 \equiv 8 \pmod{5} \Rightarrow 3 \bmod 5 = 8 \bmod 5$

Residue Classes :

- A residue class $[a]$ or $[a_n]$ is the set of integers congruent modulo 'n'.
- In other words, it is the set of all integers such that $X = a \pmod{n}$.
- A residue class $[a]$ is the set of all integers that give the same remainder when divided by n .

Example, Consider $n=5$, then

$$[0] = \{ \dots, -10, -5, 0, 5, 10, \dots \}$$

$$[1] = \{ \dots, -9, -4, 1, 6, 11, 16, \dots \}$$

$$[2] = \{ \dots, -8, -3, 2, 7, 12, 17, \dots \}$$

$$[3] = \{ \dots, -7, -2, 3, 8, 13, 18, \dots \}$$

$$[4] = \{ \dots, -11, -6, -1, 4, 9, 14, 19, \dots \}$$

- The integer in the set $[0]$ are all reduced to zero when we apply modulo 5 on them.
- In each set there is one element, the least non –ve residue. In set $[0]$, this element is 0 , in set $[1]$ it is 1 and so on.
- The set of all these least residues is what we have shown as $Z_5 = \{ 0, 1, 2, 3, 4 \}$.
- In other word, the set Z_n is the set of all least residue modulo n .

Operations in Z_n :

- The three binary operations addition, subtraction, and multiplication can also be defined for the set Z_n .

Example, 1) Add 7 to 14 in Z_{15} .

2) Subtract 11 from 7 in Z_{13} .

3) Multiply 11 from 7 in Z_{20} .

Properties of Mod operator

$$1) (a + b) \bmod n = [(a \bmod n) + (b \bmod n)] \bmod n$$

$$2) (a - b) \bmod n = [(a \bmod n) - (b \bmod n)] \bmod n$$

$$3) (a * b) \bmod n = [(a \bmod n) * (b \bmod n)] \bmod n$$

Inverses

- We are normally looking for an additive inverse or multiplicative inverse. .

1] Additive inverse:

- In Z_n two numbers a & b are additive inverse of each other if $a + b \equiv 0 \pmod{n}$
- In Z_n the additive inverse of a can be calculated as - $b = n - a$.
- For example, the additive inverse of 4 in Z_{10} is $(10-4) = 6$.

Q) Find all additive inverse pairs in Z_{10} .

Ans: $(0,0)$, $(1,9)$, $(2, 8)$, $(3, 7)$, $(4, 6)$ and $(5, 5)$

2] Multiplicative inverse:

➤ In Z_n two numbers a & b are multiplicative inverse of each other if $a * b \equiv 1 \pmod{n}$

➤ **Note:** 'a' has a multiplicative inverse in Z_n iff $\gcd(n, a) = 1$.

Q) If the modulus is 10, find the multiplicative inverse of 3 .

C.W.

Q1] Find all the additive inverse pairs in Z_5 .

Ans:

We know that, in Z_n two numbers a & b are additive inverse of each other iff

$$a+b \equiv 0 \pmod{n}$$

Again we also know that $Z_5 = \{0, 1, 2, 3, 4\}$

Now the additive inverse pairs in Z_5 are:

$(0, 0)$, since $(0+0) \bmod 5 = 0$

$(1, 4)$, since $(1+4) \bmod 5 = 0$

$(2, 3)$, since $(2+3) \bmod 5 = 0$

$(3, 2)$, since $(3+2) \bmod 5 = 0$

$(4, 1)$, since $(4+1) \bmod 5 = 0$

C.W.

Q2] Find all the multiplicative inverse pairs in Z_{10} .

Ans:

We know that, in Z_n two numbers a & b are multiplicative inverse of each other iff

$$a*b \equiv 1 \pmod{n}$$

Again we also know that $Z_{10} = \{ 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 \}$

Now the multiplicative inverse pairs in Z_{10} are:

$(1, 1)$, since $(1 * 1) \bmod 10 = 1$

$(3, 7)$, since $(3 * 7) \bmod 10 = 1$

$(7, 3)$, since $(7 * 3) \bmod 10 = 1$

$(9, 9)$, since $(9 * 9) \bmod 10 = 1$

C.W.

Q3] Find all the multiplicative inverse of each nonzero elements in Z_7 .

Ans:

Co-prime

- Also known as relatively prime.
- Two positive numbers a & b are co-prime of each other iff $\text{GCD}(a, b) = 1$.
- For example, $(10, 3)$, $(22, 13)$.

Extended Euclidean Algorithm to Find MI

➤ To find the multiplicative inverse of 'b' in Z_n , EEA works as follows:

$r_1 = n;$

$r_2 = b;$

$t_1 = 0;$

$t_2 = 1;$

while ($r_2 > 0$)

{

$q = r_1 / r_2 ;$

$r = r_1 - q * r_2 ;$

$r_1 = r_2 ;$

$r_2 = r;$

$t = t_1 - q * t_2 ;$

$t_1 = t_2 ;$

$t_2 = t;$

}

If ($r_1 == 1$) then $b^{-1} = t_1 \bmod n$.

Q] Find the multiplicative inverse of 11 in Z_{26} .

Ans:

Here, $n= 26$ and $b = 11$

Now,

q	r ₁	r ₂	r	t ₁	t ₂	t
2	26	11	4	0	1	-2
2	11	4	3	1	-2	5
1	4	3	1	-2	5	-7
2	3	1	0	5	-7	26
	1	0		-7	26	

There for $11^{-1} = - 7 \bmod 26 = 19$

```
r1 = n;  
r2 = b;  
t1 = 0;  
t2 = 1;  
while (r2 > 0)  
{  
  q = r1 / r2 ;  
  r = r1 – q * r2 ;  
  r1 = r2 ;  
  r2 = r;  
  t = t1 – q * t2 ;  
  t1 = t2 ;  
  t2 = t;  
}  
If (r1 ==1) then b-1 = t1 mod n.
```

C.W.

Q] Find the multiplicative inverse of 23 in Z_{100} .

Solⁿ:

Here, $b = 23$, $Z_n = Z_{100}$ i.e. $n = 100$

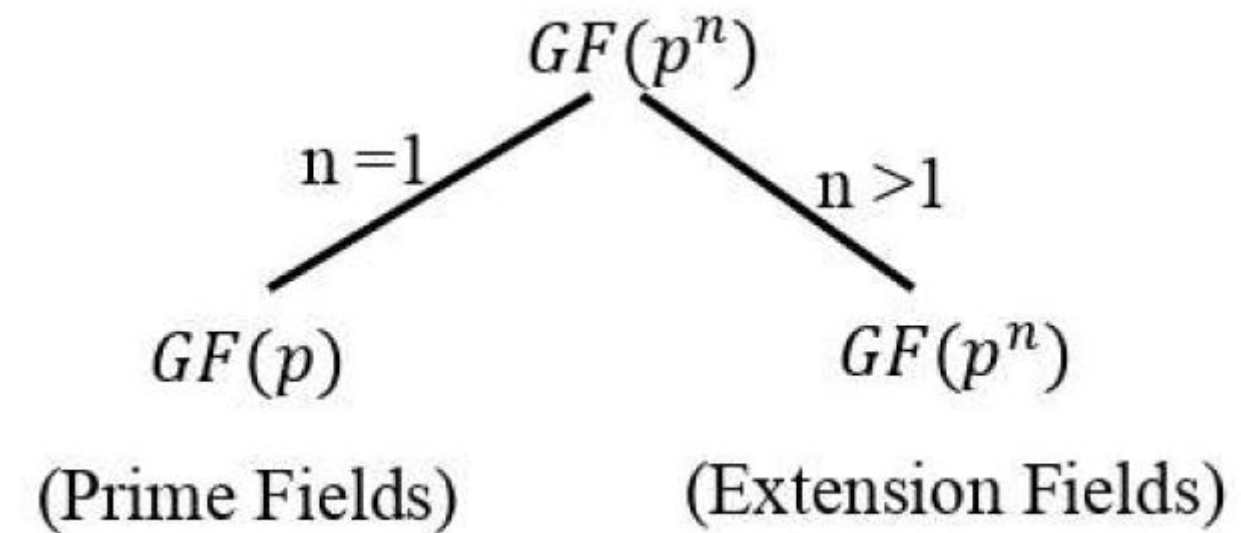
Now,

q	r_1	r_2	r	t_1	t_2	t
4	100	23	8	0	1	-4
2	23	8	7	1	-4	9
1	8	7	1	-4	9	-13
7	7	1	0	9	-13	100
	1	0		-13	100	

$$\therefore 23^{-1} = -13 \bmod 100 = 87$$

Galois Field

- A Galois field can be defined as a set of numbers that we can add, subtract, multiply and divide together and only ever end up with a result that exists in our set of numbers.
- The number of element of a Galois field is of the form P^n , where P is a prime and n is a +ve integer.
- Generally it is denote by $GF(P^n)$
- For a given prime p , $GF(p)$ is defined as the set $Z_p = \{ 0,1,2,3,\dots, p-1 \}$ of integers together with arithmetic operations modulo p .



Arithmetic in GF(7)

+	0	1	2	3	4	5	6
0	0	1	2	3	4	5	6
1	1	2	3	4	5	6	0
2	2	3	4	5	6	0	1
3	3	4	5	6	0	1	2
4	4	5	6	0	1	2	3
5	5	6	0	1	2	3	4
6	6	0	1	2	3	4	5

Arithmetic in GF(7)

X	0	1	2	3	4	5	6
0	0	0	0	0	0	0	0
1	0	1	2	3	4	5	6
2	0	2	4	6	1	3	5
3	0	3	6	2	5	1	4
4	0	4	1	5	2	6	3
5	0	5	3	1	6	4	2
6	0	6	5	4	3	2	1

Arithmetic in GF(7)

Additive and multiplicative inverse :

W	- W	W ⁻¹
0	0	---
1	6	1
2	5	4
3	4	5
4	3	2
5	2	3
6	1	6

Polynomial Arithmetic :

- A polynomial of degree n is an expression of the form-

$$f(x) = a_0 + a_1 x + a_2 x^2 + \text{-----} + a_n x^n$$

Where, x^i is called the i^{th} term and a_i is called coefficient of the i^{th} term.

- Polynomial arithmetic deals with the addition, subtraction, multiplication and division of polynomials.

Representation of 8-bit word using polynomial

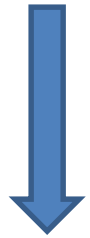
1	0	0	1	1	0	0	1
$1x^7$	$0x^6$	$0x^5$	$1x^4$	$1x^3$	$0x^2$	$0x^1$	$1x^0$

First simplification: $1x^7 + 0x^6 + 0x^5 + 1x^4 + 1x^3 + 0x^2 + 0x^1 + 1x^0$

Second simplification: $x^7 + x^4 + x^3 + 1$

Representation of polynomials in 8-bit form

$$f(x) = x^5 + x^2 + x$$



$0x^7$ $0x^6$ $1x^5$ $0x^4$ $0x^3$ $1x^2$ $1x^1$ $0x^0$

0	0	1	0	0	1	1	0
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Operation on polynomial

- Polynomial arithmetic deals with the addition, subtraction, multiplication and division of polynomials.
- Here arithmetic on the coefficients is performed modulo p ; that is, the coefficients are in $GF(P)$.
- Example: Calculate the result of the following polynomials over $GF(2)$.

- $(x^7 + x^5 + x^4 + x^3 + x + 1) + (x^3 + x + 1)$
- $(x^7 + x^5 + x^4 + x^3 + x + 1) - (x^3 + x + 1)$
- $(x^7 + x^5 + x^4 + x^3 + x + 1) \times (x^3 + x + 1)$
- $(x^7 + x^5 + x^4 + x^3 + x + 1) / (x^3 + x + 1)$

Solⁿ:

Q. Divide $5x^2 + 4x + 6$ by $2x + 1$ over $GF(7)$.

Solⁿ:

$$\begin{array}{r}
 6x + 6 \\
 2x + 1 \overline{) 5x^2 + 4x + 6} \\
 \underline{12x^2 + 6x} \\
 5x^2 + 6x \\
 \underline{-2x + 6} \\
 5x + 6 \\
 \underline{12x + 6} \\
 5x + 6 \\
 \underline{-} \\
 \times
 \end{array}$$

- 1) We must start by dividing $5x^2$ by $2x$. This requires that we divide 5 by 2 in $GF(7)$. Dividing 5 by 2 is the same as multiplying 5 by the multiplicative inverse of 2. Multiplicative inverse of 2 is 4 since $2 * 4 \bmod 7 = 1$. So we have $\frac{5}{2} = 5 * 2^{-1} = 5 * 4 = 20 \bmod 7 = 6$. Therefore, the first term of the quotient is $6x$.
- 2) Since the product of $6x$ and $2x + 1$ is $(12x^2 + 6x) \bmod 7 = 5x^2 + 6x$, we need to subtract $5x^2 + 6x$ from the dividend $5x^2 + 4x + 6$. The result is $(-2x + 6) \bmod 7 = 5x + 6$.
- 3) Our new dividend for the next round of long division is therefore $5x + 6$. To find the next quotient term, we need to divide $5x$ by the first term of the divisor that is by $2x$. Reasoning as before, we see that the next quotient term is again 6.
- 4) Applying the same process as above, we get the remainder is zero.

Cryptography

Unit- 2.4 Number Theory

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Prime number:

- A prime number is a positive integer greater than 1 that has exactly two distinct positive divisor; 1 and the number itself.
- For example, 2, 3, 5, 7, 11, 13, 17, 19 etc are prime numbers.
- Prime numbers have some unique properties -
 1. They have no divisors other than 1 and themselves.
 2. Every integer greater than 1 can be uniquely represented as a product of prime numbers known as its prime factorization. For eg, the prime factorization of 12 is $12 = 2 * 2 * 3$
 3. Prime numbers have a specific distribution pattern among the positive integer known number theorem. This theorem states that the density of prime numbers among the +ve integers decreases as the number get larger, and primes becomes less frequent as number increases.

Primality Testing

- A primality test is an algorithm to decide whether an input number is prime.

Miller-Rabin Algorithm for PT :

- Miller-Rabin algorithm is used to test the primality of large numbers.
- It works as follow-

1. Write $n-1 = 2^k \times m$, where m is odd number.

2. Choose a random number a ; $1 \leq a \leq n-1$.

3. Compute $b = a^m \bmod n$.

4. If $b \equiv 1 \pmod{n}$ then return PRIME .

5. For $i = 0$ to $k-1$

Do

 If $b \equiv -1 \pmod{n}$ then return PRIME.

 Else $b = b^2 \bmod n$

6. Return COMPOSITE.

Example

Q) Determine whether the integer 17 is prime or not by using Miller-Rabin algorithm.

Solⁿ :

Here given $n = 17$

1. $n - 1 = 16 = 2^4 \times 1$, $k = 4$ and $m = 1$
2. Let $a = 5$
3. $b = a^m \bmod n \Rightarrow b = 5^1 \bmod 17 = 5$
4. $b \equiv 1 \pmod{n} \Rightarrow 5 \equiv 1 \pmod{17} \rightarrow \text{false}$
5. So , $i = 0$ to $k-1$
It 0 : $i = 0$; $b = 5$
 $b+1 \bmod n = 0 \rightarrow 5+1 \bmod 17 = 0$; false
 $b = b^2 \bmod n \rightarrow b = 8$

1. Write $n-1 = 2^k \times m$, where m is odd number.
2. Choose a random number a ; $1 \leq a \leq n-1$.
3. Compute $b = a^m \bmod n$.
4. If $b \equiv 1 \pmod{n}$ then return PRIME .
5. For $i = 0$ to $k-1$
Do
 If $b \equiv -1 \pmod{n}$ then return PRIME.
 Else $b = b^2 \bmod n$
6. Return COMPOSITE.

It1 : $i = 1 ; b = 8;$

$b+1 \bmod n = 0 \rightarrow 8+1 \bmod 17 = 0;$ false

$b = b^2 \bmod n \rightarrow b = 8^2 \bmod 17 = 13$

It2 : $i = 2 ; b = 13;$

$b+1 \bmod n = 0 \rightarrow 13+1 \bmod 17 = 0;$ false

$b = b^2 \bmod n \rightarrow b = 13^2 \bmod 17 = 16$

It3 : $i = 3 ; b = 16;$

$b+1 \bmod n = 0 \rightarrow 16+1 \bmod 17 = 0;$ True

Return PRIME

Therefore the integer 17 is a prime number.

1. Write $n-1 = 2^k \times m$, where m is odd number.
2. Choose a random number a ; $1 \leq a \leq n-1$.
3. Compute $b = a^m \bmod n$.
4. If $b \equiv 1 \pmod{n}$ then return PRIME.
5. For $i = 0$ to $k-1$
Do
 If $b \equiv -1 \pmod{n}$ then return PRIME.
 Else $b = b^2 \bmod n$
6. Return COMPOSITE.

Q) Determine whether the given integers are prime or not by using Miller-Rabin algorithm.

a) 7

b) 10

c) 101

Fermat's Little Theorem

- Fermat's theorem states that “ If P is a prime & b is a +ve integer (not divisible by P) then,

$$b^{P-1} \equiv 1 \pmod{P}$$

Where $\gcd(b, P) = 1$

- An alternative form of Fermat's theorem is –

$$b^P \equiv b \pmod{P}$$

- Example: let $P=7$ and $b=3$

$$b^{P-1} \equiv 1 \pmod{P} \rightarrow 3^6 \equiv 1 \pmod{7}$$

$$\rightarrow 729 \pmod{7} = 1 \pmod{7}$$

$$\rightarrow 1 = 1$$

Euler Totient Function ($\phi(n)$) :

- Euler totient function (also called phi-function) counts the number of positive integers less than 'n' that are co-prime to 'n'.

Example : $\phi(10) = ?$

Here, $n = 10$

set of residue = { 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 }

reduced set of residue = { 1, 3, 7, 9 } these are the number that are relatively prime to 10.

Therefore

$$\phi(10) = 4$$

- If 'p' is a prime then $\phi(p) = p-1$
- If 'p' & 'q' are both prime then $\phi(pq) = (p-1)(q-1)$

Example : $\phi(21) = ?$

Euler Theorem :

- Euler's theorem states that if 'a' and 'n' are co-prime positive integers then,
$$a^{\phi(n)} \equiv 1 \pmod{n}$$

Where $\phi(n)$ is a Euler totient function.

- For Example: let $a = 3$; $n = 10$

Here $\phi(10) = \{1, 3, 7, 9\} = 4$

Now

$$a^{\phi(n)} \equiv 1 \pmod{n}$$

$$3^4 \equiv 1 \pmod{10}$$

$$81 \bmod 10 = 1 \bmod 10$$

$$1 = 1$$

Primitive Root

- Integer 'a' is said to be a primitive root of prime number 'p' if $a^1 \bmod p$, $a^2 \bmod p$, $a^3 \bmod p$, , $a^{p-1} \bmod p$ are distinct and consists of integers from 1 to p-1 in permutation.
- Example: Is 2 a primitive root of 5 ?

Solⁿ:

Here, $a = 2$ and $p = 5$

$$2^1 \bmod 5 = 2$$

$$2^2 \bmod 5 = 4$$

$$2^3 \bmod 5 = 3$$

$$2^4 \bmod 5 = 1$$

Here, all values are distinct and consists of integers 1 to 4.

$\therefore$ 2 is primitive root of 5.

Primitive Root

➤ Example: Is 2 a primitive root of 7 ?

Solⁿ:

Here, $a = 2$ and $p = 7$

$$\begin{array}{l} 2^1 \bmod 7 = 2 \\ 2^2 \bmod 7 = 4 \\ 2^3 \bmod 7 = 1 \\ 2^4 \bmod 7 = 2 \end{array} \quad \begin{array}{c} \curvearrowleft \\ \text{Repeat} \\ \curvearrowright \end{array}$$

$\therefore 2$ is not primitive root of 7.

!!!Thank you!!!